



CADEN TOH

ASPIRING MACHINE LEARNING ENGINEER

Year 2 Diploma in Applied AI & Analytics student at Singapore Polytechnic passionate about Machine Learning, AI and Data Engineering. Experienced in developing AI solutions using Python, TensorFlow, Scikit-learn and SQL through academic and hackathon projects. Seeking to contribute to real-world AI and smart manufacturing solutions.



CONTACT



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Singapore



TECHNICAL SKILLS

PROGRAMMING

Python • SQL • JavaScript
HTML • CSS

MACHINE LEARNING

TensorFlow • Keras • Scikit-learn
CNN • DNN • RNN • NLP

DATA ANALYTICS

Pandas • NumPy • Power BI

BACKEND DEVELOPMENT

Node.js • Express.js • REST APIs
MySQL

TOOLS

Git • GitHub • VS Code



CERTIFICATIONS

- Supervised Learning with Scikit-learn
- Standard First Aid, CPR & AED Provider Course



EDUCATION

Singapore Polytechnic
Diploma in Applied AI & Analytics
Apr 2025 – Mar 2028

Relevant Coursework (Distinctions)

- ✓ AI & Machine Learning
- ✓ Backend Web Development
- ✓ Business Statistics
- ✓ Fundamentals of Computing
- ✓ Mathematics



PROJECTS



CNN Image Classification with EMNIST Letters

Python • TensorFlow • Keras • NumPy • Matplotlib

- Built and trained a CNN model to classify letters from the EMNIST dataset.
- Achieved 94.28% test accuracy and improved from a baseline of 91.20%.
- Applied data preprocessing, augmentation and hyperparameter tuning to enhance model performance.



Variational Autoencoder (VAE) on Fashion MNIST

Python • TensorFlow • Keras • UMAP • Matplotlib

- Implemented and trained a VAE to learn underlying latent representations.
- Experimented with different latent dimensions and visualized latent space using UMAP.
- Evaluated generative quality using reconstruction loss and FID scores.



AI Burnout Support Chatbot

Python • OpenAI API • NLP • Prompt Engineering

- Developed an AI-powered chatbot that provides personalised burnout support.
- Integrated OpenAI API and designed effective prompts for contextual, empathetic conversations.
- Focused on natural language understanding and user experience.



Power BI Dashboard – Resident Data Analysis

Power BI • DAX • Data Visualisation

- Analyzed resident dataset to uncover insights on engagement, age distribution and MRT accessibility.
- Built interactive dashboards with KPIs, ranked charts and scatter plots.
- Applied DAX calculations and data transformations for accurate insights.



SQL ETL & Data Warehouse Project

MySQL • SQL • Data Modeling

- Cleaned raw data and performed ETL processes to build a star schema.
- Designed and created fact & dimension tables for analytical reporting.
- Wrote complex SQL queries to extract insights from data warehouse.



FinTech Hackathon with Prudential – Gamified Insurance Platform

React • HTML • CSS • JavaScript • Design Thinking

- Collaborated in a team to develop a simple gamified insurance platform aimed at improving financial literacy and engagement among Gen Z users.
- Conducted user research, ideated features and built a functional prototype using React, HTML and CSS.
- Presented our solution to industry mentors and received valuable feedback.



ADDITIONAL ACTIVITIES



Singapore Poly Lifesavers – Vice Captain

- Led and supported training sessions and team operations.
- Assisted in planning activities and mentoring junior members.



National Day Parade 2023 Participant

- Represented Singapore Polytechnic in the National Day Parade 2023.



National Lifesaving Championship 2026

- Represented Singapore Polytechnic.
- Achieved Gold Medal in the National Lifesaving Championship.



INTERESTS



Machine Learning



Data Engineering



Computer Vision



Smart Manufacturing



Artificial Intelligence



Automation & Robotics